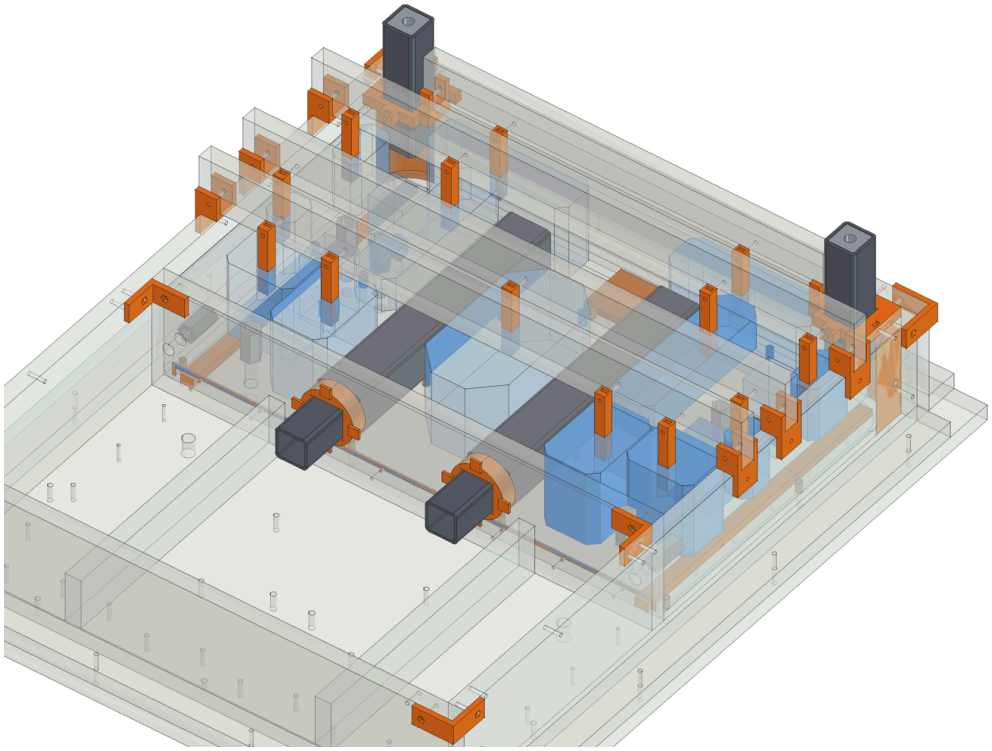
Base mold

DIY CNC epoxy-granite mill · October 2026

How to make, put together and take apart the mold for the base: both halves, cast one after the other on one floor board. The first section (how the molds work) and the last three (pouring, taking it apart, fixes) are the same in every mold guide; molds-complete.pdf has all the molds and the sheet layout for the boards.

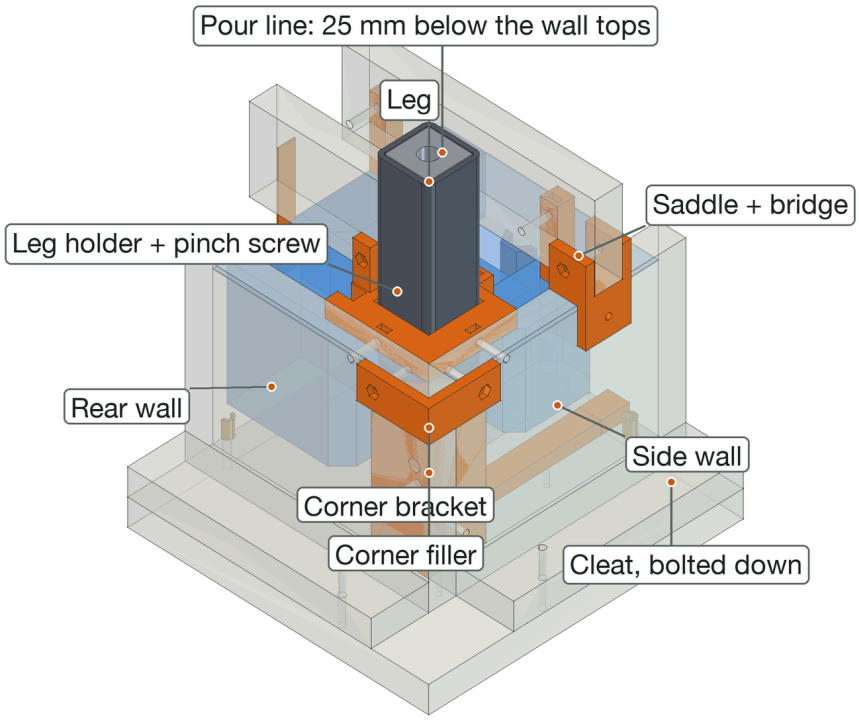
The build guide covers the casting itself (mix, pour, cure); this guide is everything before and after that. All the sizes and hole positions come from the model (`model/build_cnc.py`), so if you change the design, re-run it and the guides, hole maps and printed parts all follow.



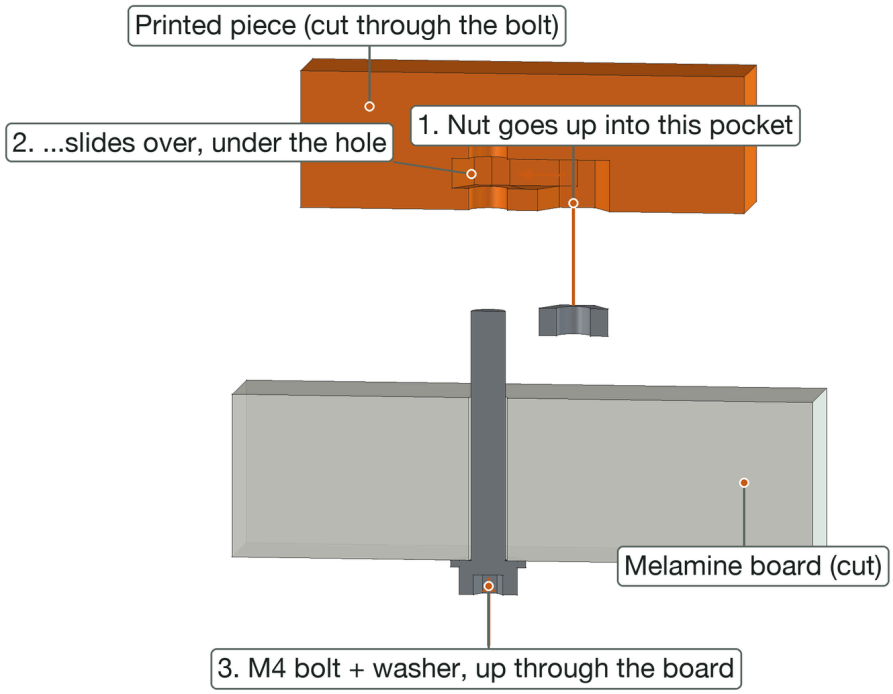
The order of work: print the kit (start first: it is days of printing) → cut and drill the boards → cut the foam → wax → assemble the mold → cast → take it apart. The castings go in this order: base rear half, base front half (against the rear half), the two uprights together, the beam.

1. How the molds work

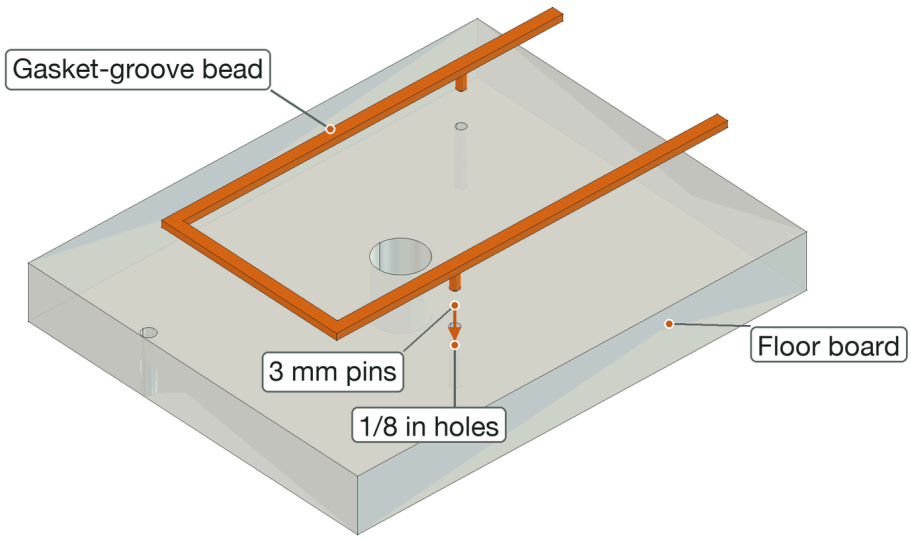
Walls stand 25 mm above the pour line. All the bolts that hold the box together go through that band, so none passes through a face that gets cast against. Fill to the line, not to the top. Strips of board (cleats) bolted down along the outside of each wall's foot keep the walls from moving out; printed corner brackets hold the corners square at the top.



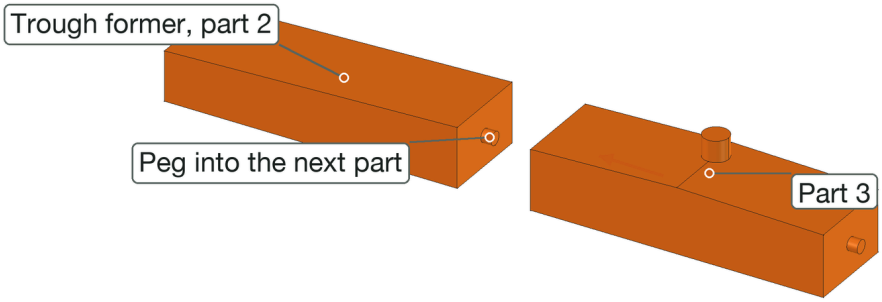
Printed pieces bolt on from outside. Each piece that forms part of the casting's surface (the formers and fillers) has a nut built into it: the nut drops into a pocket in the face that sits against the board, slides sideways under the bolt hole, and an M4 bolt (M3 on thin pieces) goes through the board from outside into it. The pocket is on the board side, so nothing opens onto a cast face, and the bolt comes out from outside after casting. Fit the nuts before the pieces go on.



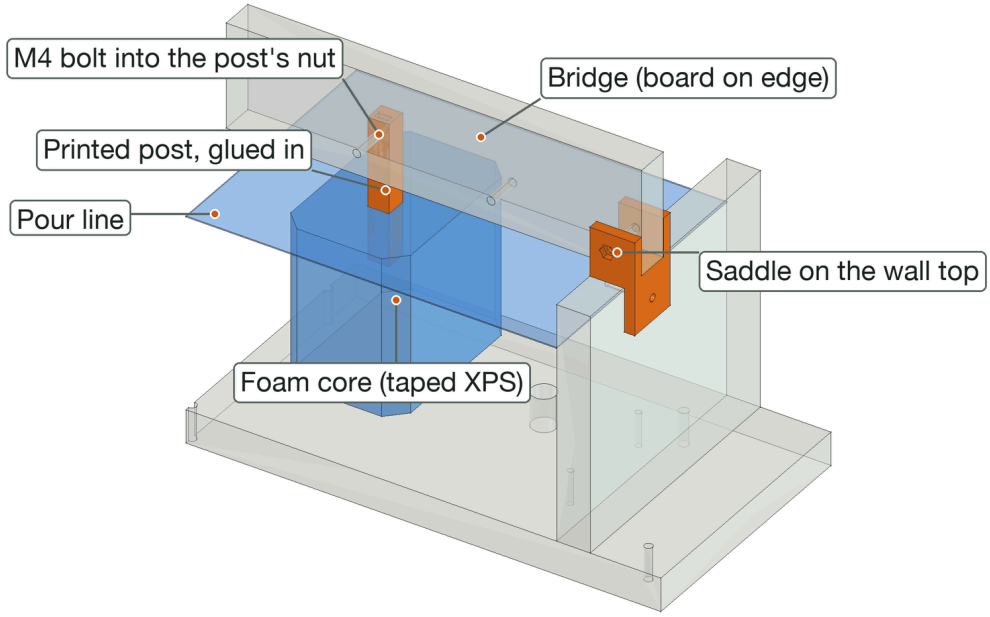
Thin pieces have pins. The gasket-groove beads and the motor relief are too thin for a nut; their 3 mm pins push into 1/8 in holes. The casting presses them onto the board.



Pieces longer than the printer bed are printed in parts with a peg on one cut face and a socket in the other. Push them together; each part has its own bolts to the board, so they need no glue.



Cores (the pockets in the base underside) are blocks of XPS foam wrapped in packing tape. Each has a printed post glued into it; the post bolts to the side of a bridge (a board standing on edge across the mold, in printed saddles on the wall tops), which sets its height and stops it floating.



Fits are 0.15 mm a side: snug, but they go together by hand. Board thickness matters: the saddles and cradles are drawn for 19.05 mm board (Home Depot's 3/4 in white melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 19 mm (3/4 in)	about 1.5 m2	all the molds come out of two 4 x 8 ft sheets (layout in molds-complete.pdf)
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores; stack two layers for the 90 mm ones
PETG	0.73 kg	this mold's printed parts (sliced); all the molds and machine parts fit in two 1 kg rolls
Packing tape, double-sided carpet tape	1 roll each	foam wrapping and holding
Paste wax (Partall #2) and PVA	1 each	release, 3-5 coats
3/8 in quarter-round molding (PVC)	a few meters	the 10 mm outside edges of the castings

Hardware (zinc-plated is fine; grease anything that touches the mix). Buy about 10 % spare of the small bolts, nuts and washers.

Item	Qty
M3 nut	20
M3 washer	20
M3 x 25 bolt	20
M4 nut	112
M4 washer	108
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	68
M4 x 35 bolt	24
M4 x 45 bolt	16
M5 x 20 screw, greased (holds an insert)	21
M6 x 20 screw, greased (holds an insert)	6
1/2-13 x 1-1/2 in bolt, greased (holds a draw-rod coupler)	2
1/2-13 x 2-1/2 in bolt, greased (holds a coupler)	8

Tools: a drill with sharp brad-point bits (sizes in section 5) and a 2-1/2 in hole saw; a square and a long straightedge; a few clamps; a 7 mm nut driver and a 3 mm hex key; a fine saw or a hot-wire cutter and a sanding block for the foam; a marker that writes on melamine.

3. Print the parts

80 prints, 0.73 kg of PETG and about 65 hours of printing, sliced in Bambu Studio for an A1 mini (settings in the print guide). Print one corner bracket and one saddle first and check them on an offcut of your board: the saddle should drop over it with a little play.

Folder	File	Qty	g (all)	Time each
base-box	`kit-corner-bracket.stl`	4	33	39m 28s
base-box	`kit-partition-bracket.stl`	2	14	35m 28s
base-box	`kit-saddle.stl`	6	58	46m 3s
base-box	`kit-leg-holder.stl`	4	83	1h 37m 19s
base-rear	`former-gutter-L-part1of3.stl`	1	8	35m 5s
base-rear	`former-gutter-L-part2of3.stl`	1	8	33m 40s
base-rear	`former-gutter-L-part3of3.stl`	1	7	31m 15s
base-rear	`former-gutter-R-part1of3.stl`	1	8	35m 6s
base-rear	`former-gutter-R-part2of3.stl`	1	8	33m 40s
base-rear	`former-gutter-R-part3of3.stl`	1	7	31m 15s
base-rear	`former-motor-relief.stl`	1	12	40m 58s
base-rear	`filler-corner-L.stl`	1	17	1h 44m 24s
base-rear	`filler-corner-R.stl`	1	16	1h 42m 27s

Folder	File	Qty	g (all)	Time each
base-front	`former-gutter-L-part1of2.stl`	1	11	49m 23s
base-front	`former-gutter-L-part2of2.stl`	1	10	43m 43s
base-front	`former-gutter-R-part1of2.stl`	1	11	49m 24s
base-front	`former-gutter-R-part2of2.stl`	1	10	43m 43s
base-front	`former-trough-part1of5.stl`	1	20	1h 14m 18s
base-front	`former-trough-part2of5.stl`	1	22	1h 23m 56s
base-front	`former-trough-part3of5.stl`	1	25	1h 33m 33s
base-front	`former-trough-part4of5.stl`	1	23	1h 24m 3s
base-front	`former-trough-part5of5.stl`	1	21	1h 17m 36s
base-front	`filler-corner-L.stl`	1	21	2h 0m 50s
base-front	`filler-corner-R.stl`	1	21	1h 56m 31s
base-rear	`former-split-gasket-groove-part1of4.stl`	1	2	9m 50s
base-rear	`former-split-gasket-groove-part2of4.stl`	1	1	8m 58s
base-rear	`former-split-gasket-groove-part3of4.stl`	1	1	8m 58s
base-rear	`former-split-gasket-groove-part4of4.stl`	1	2	9m 53s
base-front	`core-center-post.stl`	1	5	44m 56s
base-rear	`core-center-post.stl`	1	5	44m 56s
base-front	`core-outboard-L-post.stl`	1	5	44m 53s
base-front	`core-trough-band-L-post.stl`	1	5	44m 54s
base-front	`core-draw-rod-L-post.stl`	1	5	44m 53s
base-front	`core-foot-a-L-post.stl`	1	5	44m 53s
base-rear	`core-foot-b-L-post.stl`	1	5	44m 53s
base-rear	`core-foot-c-L-post.stl`	1	5	44m 53s
base-rear	`core-knee-L-post.stl`	1	5	44m 53s
base-rear	`core-outer-a-L-post.stl`	1	5	44m 53s
base-rear	`core-outer-b-L-post.stl`	1	5	44m 53s
base-rear	`core-rear-L-post.stl`	1	5	44m 53s
base-front	`core-outboard-R-post.stl`	1	5	44m 53s
base-front	`core-trough-band-R-post.stl`	1	5	44m 54s
base-front	`core-draw-rod-R-post.stl`	1	5	44m 53s
base-front	`core-foot-a-R-post.stl`	1	5	44m 53s
base-rear	`core-foot-b-R-post.stl`	1	5	44m 53s
base-rear	`core-foot-c-R-post.stl`	1	5	44m 53s
base-rear	`core-knee-R-post.stl`	1	5	44m 53s
base-rear	`core-outer-a-R-post.stl`	1	5	44m 53s

Folder	File	Qty	g (all)	Time each
base-rear	`core-outer-b-R-post.stl`	1	5	44m 53s
base-rear	`core-rear-R-post.stl`	1	5	44m 53s
base-rear	`plug-socket-spike.stl`	4	8	13m 26s
base-rear	`kit-spigot-bushing.stl`	2	34	1h 11m 51s
base-front	`cast-in-base-front-tube-plug-l.stl`	2	13	29m 7s
base-front	`cast-in-base-front-spigot-sleeve-l.stl`	2	29	1h 14m 40s
base-front	`cast-in-leg-front-l-plug.stl`	2	7	15m 49s
base-rear	`cast-in-base-rear-tube-plug-l.stl`	2	13	27m 49s
base-rear	`cast-in-base-rear-spigot-sleeve-l.stl`	2	29	1h 14m 38s
base-rear	`cast-in-leg-rear-l-plug.stl`	2	7	15m 49s

- **Nuts:** press M4 nuts into the hex pockets of the brackets and saddles; slide the nuts into the pockets of the formers and fillers (a dab of grease keeps them from falling out). A nut on a bolt through the hole pulls a tight one into place.
- **Pegs:** test-fit the parts of each long former; a light pass with a file if a peg is tight.

4. Cut the boards

Most boards are strips, so the quickest way is to have the store rip each sheet into strips at the widths below, and crosscut them at home. Label every board with its name on the outside face as you cut it, and mark which face is the cast face (the melamine face that the mix touches).

Board	Qty	Size (mm)	Notes
Base floor	1	718 x 843	both halves are cast on it
Base front wall	1	638 x 135	
Base rear wall	1	638 x 135	
Base side wall, left	1	724 x 135	
Base side wall, right	1	724 x 135	
Base partition	1	600 x 135	rear half only; comes out before the front half
Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base cleat, front	1	638 x 40	offcut strip, flat on the floor
Base cleat, rear	1	638 x 40	offcut strip, flat on the floor
Base cleat, left	1	724 x 40	offcut strip, flat on the floor
Base cleat, right	1	724 x 40	offcut strip, flat on the floor
Base bridge y=-190	1	651 x 60	
Base bridge y=-58	1	651 x 60	
Base bridge y=100	1	651 x 60	
Base bridge y=170	1	651 x 60	

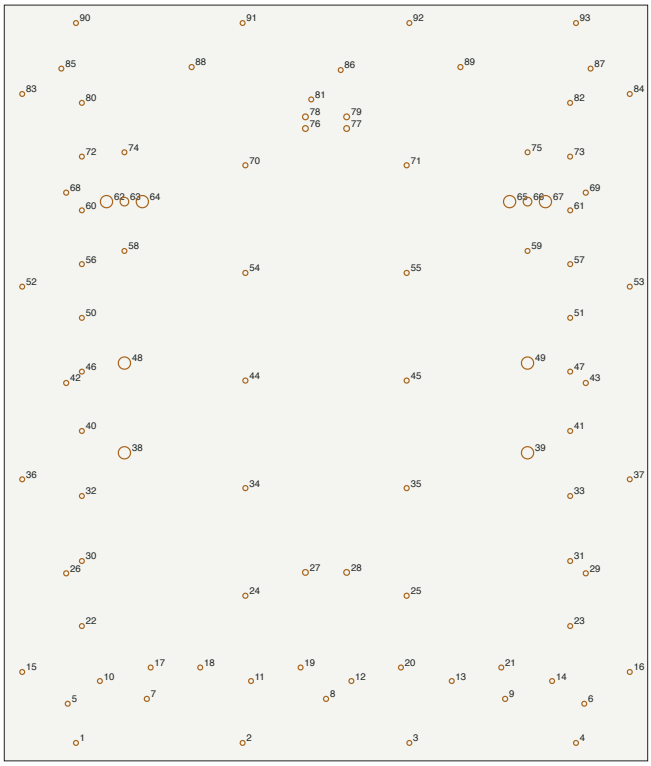
Board	Qty	Size (mm)	Notes
Base bridge y=280	1	651 x 60	
Base leg bridge y=-310	1	480 x 60	
Base leg bridge y=344	1	480 x 60	

5. Drill the boards

- Drill every hole from the **cast side**, so any tear-out is on the outside. Clamp a scrap under the board where the bit comes out.
- The drawings and tables are seen from the cast side: x from the left edge, y from the bottom edge. For a wall the bottom edge sits on the floor; for a floor, the bottom of the drawing is toward the front wall (the beam's bottom wall). Left and right are as you look at the cast face, so a floor is mirrored compared with the machine. Bridges and cleats are drawn from the top.
- The brackets and saddles are drill guides: clamp one in place and drill through its holes.
- A washer goes under every bolt head; countersink nothing.

Hole	Drill	For
3.2 mm	1/8 in	motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	cleat, core post, corner bracket, corner-L, corner-R, gutter-L, gutter-R, leg holder, saddle, socket spike, trough
5.5 mm	7/32 in	M5 insert screw
6.5 mm	17/64 in	M6 insert screw
9.6 mm	25/64 in	3/8 dowel
13.5 mm	17/32 in	1/2-13 bolt into coupler
63.5 mm	2-1/2 in hole saw	hole saw

Base floor



0,0 · 718 x 843 mm · seen from the cast side

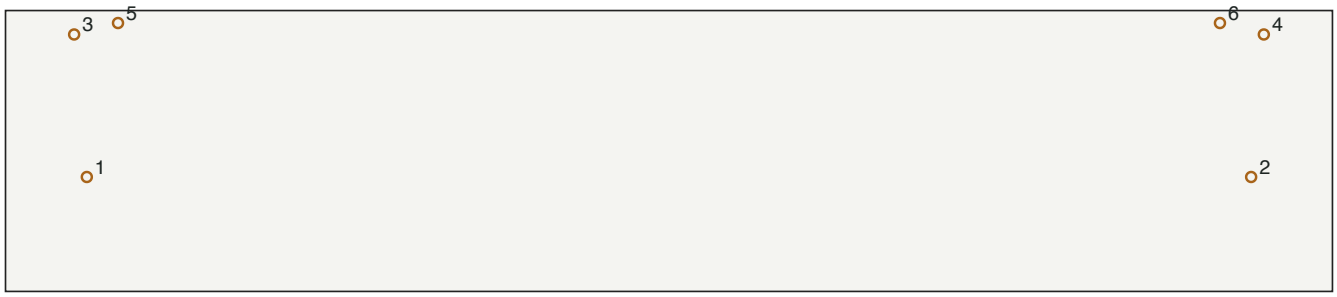
#	x	y	Hole	For
1	80	20	4.5	cleat
2	266	20	4.5	cleat
3	452	20	4.5	cleat
4	638	20	4.5	cleat
5	71	64	3.4	corner-R
6	647	64	3.4	corner-L
7	159	69	5.5	M5 insert screw
8	359	69	5.5	M5 insert screw
9	559	69	5.5	M5 insert screw
10	107	89	4.5	trough
11	275	89	4.5	trough
12	387	89	4.5	trough
13	499	89	4.5	trough
14	611	89	4.5	trough
15	20	99	4.5	cleat
16	698	99	4.5	cleat
17	163	104	4.5	trough
18	219	104	4.5	trough
19	331	104	4.5	trough

#	x	y	Hole	For
20	443	104	4.5	trough
21	555	104	4.5	trough
22	87	150	4.5	gutter-R
23	632	150	4.5	gutter-L
24	269	184	5.5	M5 insert screw
25	449	184	5.5	M5 insert screw
26	69	209	5.5	M5 insert screw
27	336	210	6.5	M6 insert screw
28	382	210	6.5	M6 insert screw
29	649	209	5.5	M5 insert screw
30	87	223	4.5	gutter-R
31	632	223	4.5	gutter-L
32	87	295	3.4	gutter-R
33	632	295	3.4	gutter-L
34	269	304	5.5	M5 insert screw
35	449	304	5.5	M5 insert screw
36	20	314	4.5	cleat
37	698	314	4.5	cleat
38	134	344	13.5	1/2-13 bolt into coupler
39	584	344	13.5	1/2-13 bolt into coupler
40	87	368	3.4	gutter-R
41	632	368	3.4	gutter-L
42	69	421	5.5	M5 insert screw
43	649	421	5.5	M5 insert screw
44	269	424	5.5	M5 insert screw
45	449	424	5.5	M5 insert screw
46	87	434	3.4	gutter-R
47	632	434	3.4	gutter-L
48	134	444	13.5	1/2-13 bolt into coupler
49	584	444	13.5	1/2-13 bolt into coupler
50	87	494	3.4	gutter-R
51	632	494	3.4	gutter-L
52	20	529	4.5	cleat
53	698	529	4.5	cleat
54	269	544	5.5	M5 insert screw

#	x	y	Hole	For
55	449	544	5.5	M5 insert screw
56	87	554	3.4	gutter-R
57	632	554	3.4	gutter-L
58	134	569	4.5	socket spike
59	584	569	4.5	socket spike
60	87	614	3.4	gutter-R
61	632	614	3.4	gutter-L
62	114	624	13.5	1/2-13 bolt into coupler
63	134	624	9.6	3/8 dowel
64	154	624	13.5	1/2-13 bolt into coupler
65	564	624	13.5	1/2-13 bolt into coupler
66	584	624	9.6	3/8 dowel
67	604	624	13.5	1/2-13 bolt into coupler
68	69	634	5.5	M5 insert screw
69	649	634	5.5	M5 insert screw
70	269	664	5.5	M5 insert screw
71	449	664	5.5	M5 insert screw
72	87	674	3.4	gutter-R
73	632	674	3.4	gutter-L
74	134	679	4.5	socket spike
75	584	679	4.5	socket spike
76	336	705	6.5	M6 insert screw
77	382	705	6.5	M6 insert screw
78	336	718	6.5	M6 insert screw
79	382	718	6.5	M6 insert screw
80	87	734	3.4	gutter-R
81	343	737	3.2	motor-relief
82	632	734	3.4	gutter-L
83	20	744	4.5	cleat
84	698	744	4.5	cleat
85	64	772	3.4	corner-R
86	375	770	3.2	motor-relief
87	654	772	3.4	corner-L
88	209	774	5.5	M5 insert screw
89	509	774	5.5	M5 insert screw

#	x	y	Hole	For
90	80	823	4.5	cleat
91	266	823	4.5	cleat
92	452	823	4.5	cleat
93	638	823	4.5	cleat

Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	39	55	4.5	corner-L
2	599	55	4.5	corner-R
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

Base rear wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	26	56	4.5	corner-R
2	612	56	4.5	corner-L
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket

#	x	y	Hole	For
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

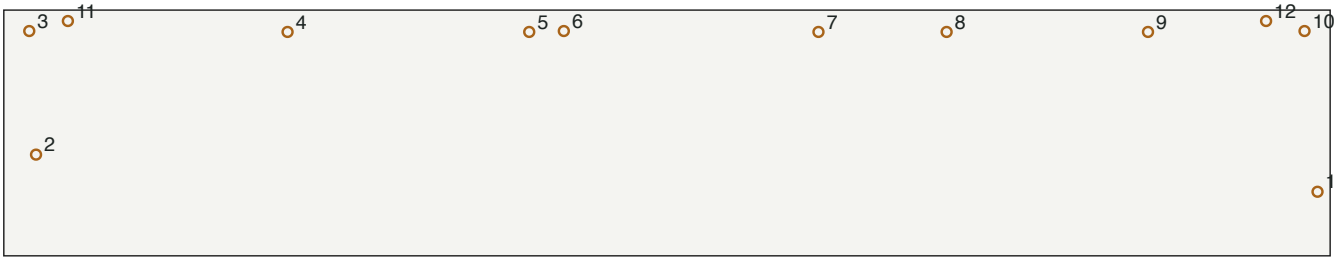
Base side wall, left



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	707	54	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	438	123	4.5	saddle
9	570	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base side wall, right

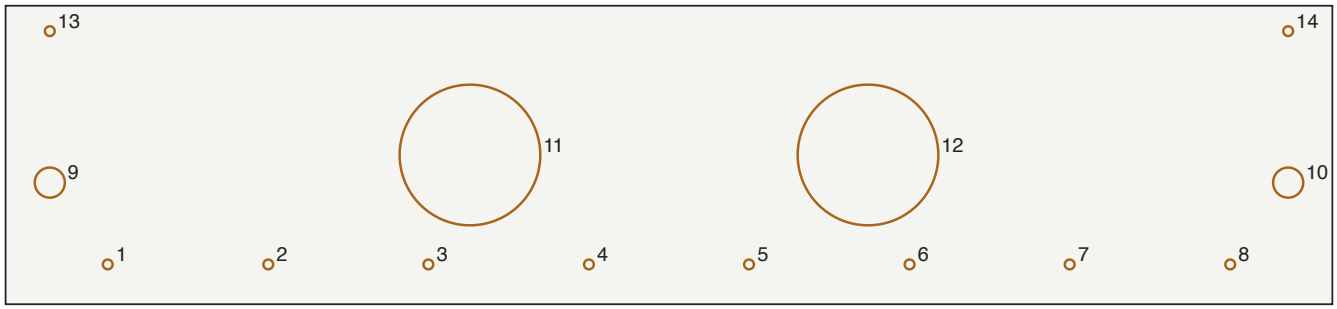


0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	718	35	4.5	corner-R

#	x	y	Hole	For
2	18	56	4.5	corner-R
3	14	124	4.5	corner bracket
4	155	123	4.5	saddle
5	287	123	4.5	saddle
6	306	124	4.5	corner bracket
7	445	123	4.5	saddle
8	515	123	4.5	saddle
9	625	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base partition

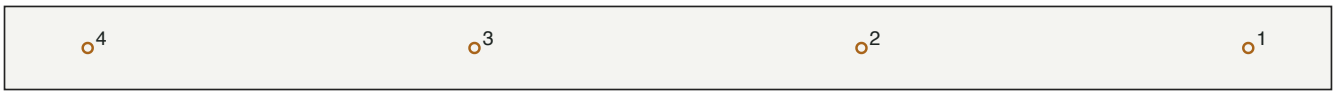


0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	20	55	13.5	1/2-13 bolt into coupler
10	580	55	13.5	1/2-13 bolt into coupler
11	210	68	63.5	hole saw
12	390	68	63.5	hole saw
13	20	124	4.5	corner bracket

#	x	y	Hole	For
14	580	124	4.5	corner bracket

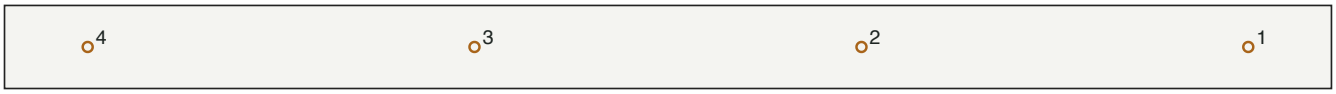
Base cleat, front



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

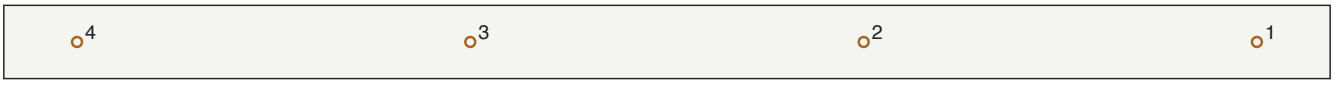
Base cleat, rear



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

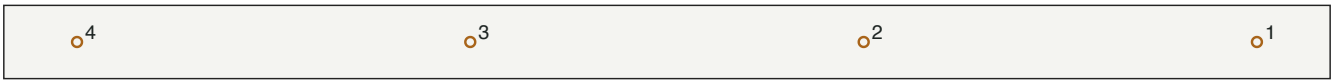
Base cleat, left



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

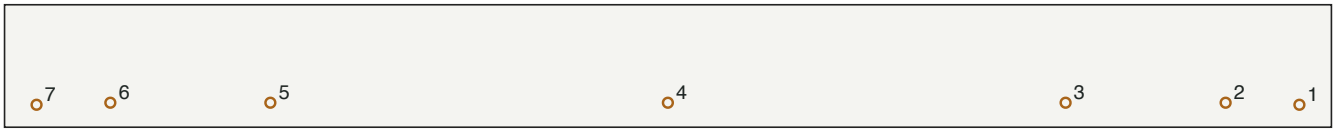
Base cleat, right



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

Base bridge y=-190



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	599	12	4.5	core post
3	520	12	4.5	core post
4	325	12	4.5	core post
5	130	12	4.5	core post
6	52	12	4.5	core post
7	16	11	4.5	saddle

Base bridge y=-58



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	489	12	4.5	core post
3	161	12	4.5	core post
4	16	11	4.5	saddle

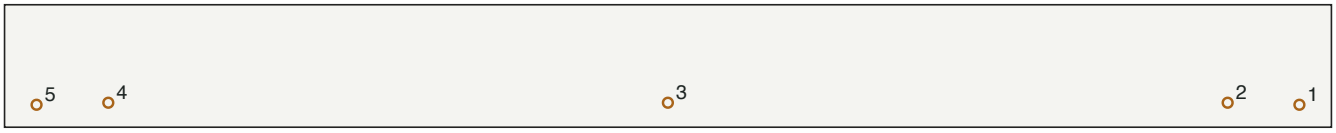
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	565	12	4.5	core post
3	489	12	4.5	core post
4	161	12	4.5	core post
5	85	12	4.5	core post
6	16	11	4.5	saddle

Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	325	12	4.5	core post
4	51	12	4.5	core post
5	16	11	4.5	saddle

Base bridge y=280

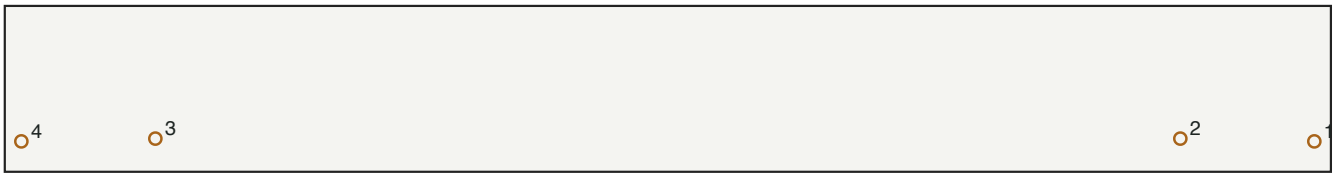


0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	480	12	4.5	core post
4	170	12	4.5	core post

#	x	y	Hole	For
5	51	12	4.5	core post
6	16	11	4.5	saddle

Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344



0,0 · 480 x 60 mm · seen from the outside (top)

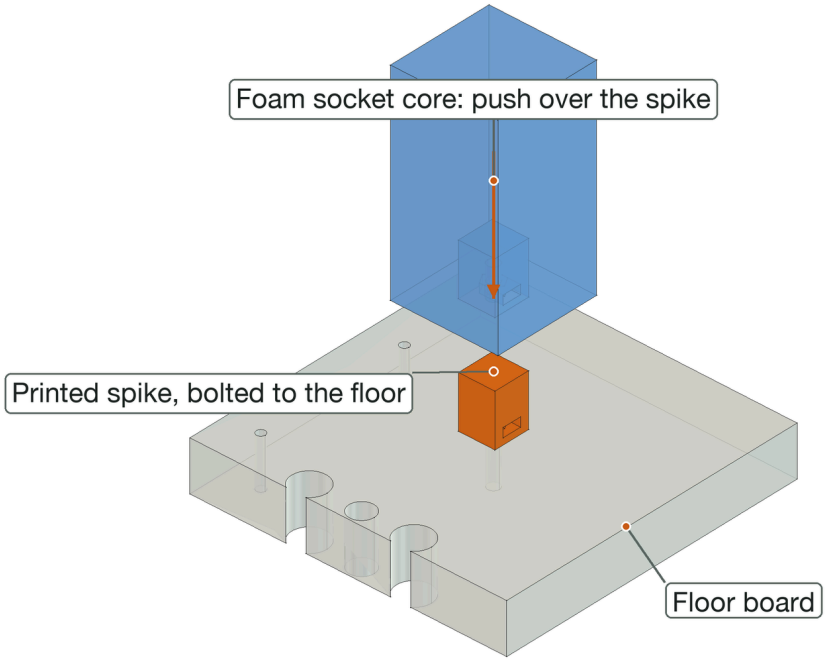
#	x	y	Hole	For
1	474	11	4.5	leg holder
2	385	12	4.5	core post
3	95	12	4.5	core post
4	6	11	4.5	leg holder

6. Cut the foam

Cut with a fine saw, a hot wire, or a long sharp knife and a sanding block. Wrap each piece in packing tape (smooth, no wrinkles: the tape is the cast face). For the 90 mm cores glue two layers of 2 in foam with a foam-safe adhesive (not solvent cement) or double-sided tape, then trim. Make each pocket core's post hole a snug 12 mm square, 30 mm deep, and glue the printed post in.

Casting	Piece	Qty	Size and how
base-front	center	1	XPS block 100 x 185 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	center	1	XPS block 100 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-L	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-L	1	XPS block 37 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-L	1	XPS block 68 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 68 x 100 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-c-L	1	XPS block 80 x 63 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 50 x 183 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-L	1	XPS block 35 x 43 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-L	1	XPS block 35 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-L	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-R	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-R	1	XPS block 37 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-R	1	XPS block 68 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-R	1	XPS block 68 x 100 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-c-R	1	XPS block 80 x 63 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-R	1	XPS block 50 x 183 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-R	1	XPS block 35 x 43 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-R	1	XPS block 35 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.

Casting	Piece	Qty	Size and how
base-rear	rear-R	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	socket	4	XPS block 41 x 41 x 80 mm, taped; dig it out after curing (a little acetone dissolves the rest).



7. Prepare the surfaces

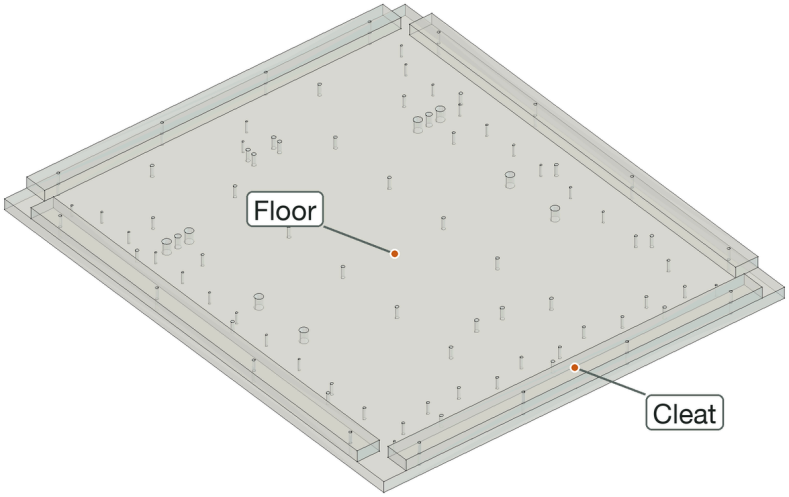
- Wax every face the mix touches, boards and removable printed pieces alike: 3-5 thin coats of paste wax, each buffed, then a coat of PVA. Don't wax the pieces that stay in (core posts, cast-in parts): scuff them and wipe them with acetone so the epoxy grips.
- The castings' outside edges are rounded 10 mm: glue 3/8 in quarter-round molding into the inside corners that form outside edges (floor to wall), and run a small silicone fillet into the others.
- Grease every screw, bolt and rod that will come out of the casting.

8. Put the mold together

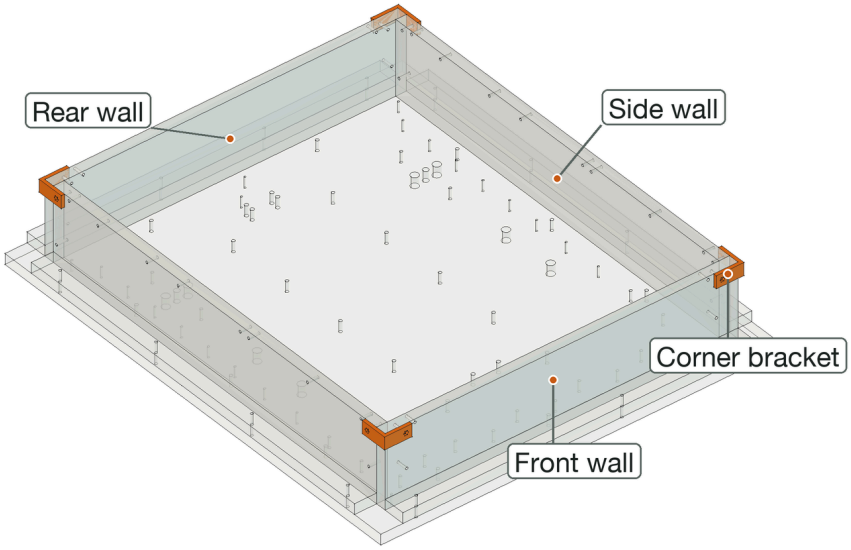
Base, rear half (cast first)

Both base halves are cast on the same floor board. The rear half goes first, against a partition at the split; then the partition comes out and the front half is cast against the rear half's waxed split face. The two top faces come out flush, the split faces match, and the front tubes slide straight onto the rear half's spigots. The base is cast upside down (its top face on the floor); these pictures show the mold the right way up.

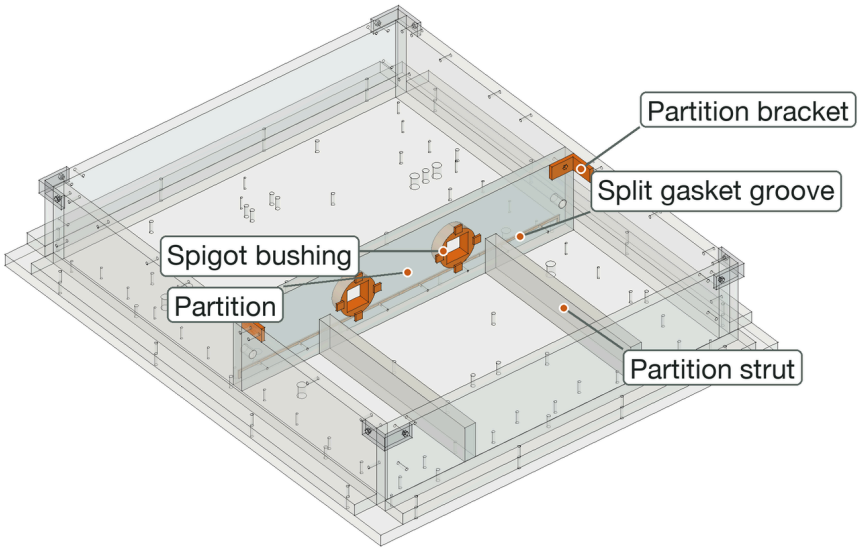
1. Stand the floor on three battens (the insert screws and floor bolts come up from underneath) and bolt the cleats down along its edges.



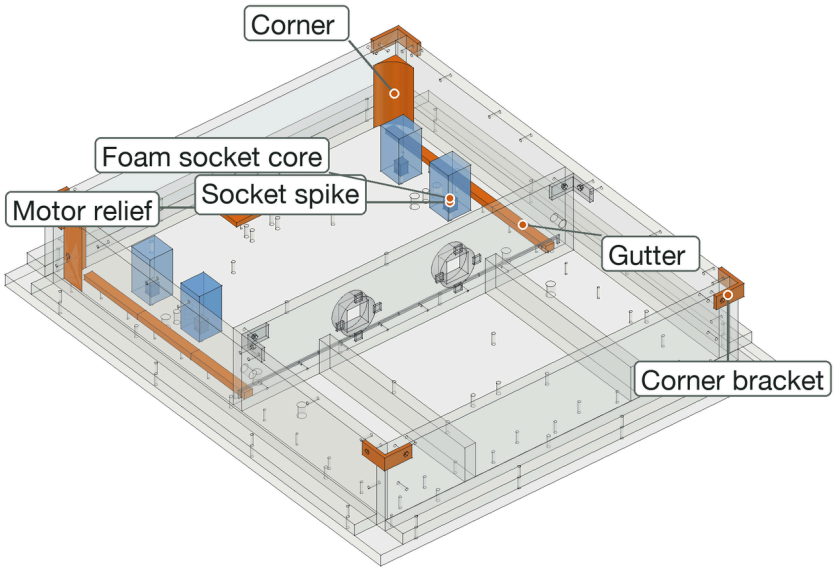
2. Stand the walls on the floor against the cleats and fit the four corner brackets at the top corners. Measure the diagonals before you tighten: they should match within 2 mm.



3. Fit the partition between the side walls with its two brackets on the front side, and the two struts from the front wall to its foot. Push the gasket-groove beads onto its pins and the spigot bushings into its holes from the front.

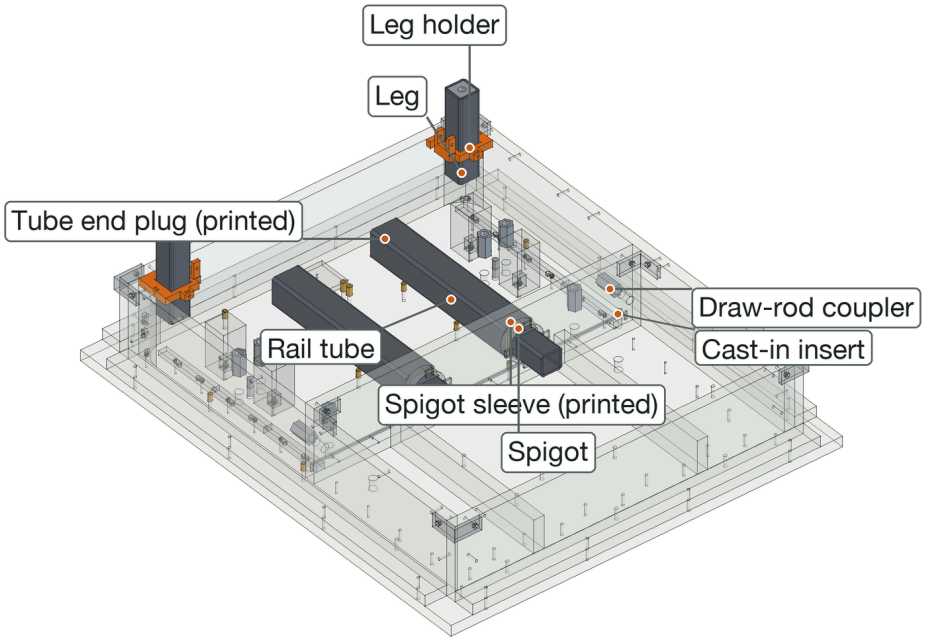


4. Bolt on the gutter formers, the motor relief (pins) and the rear corner fillers; bolt the socket spikes to the floor and push a taped foam block over each.

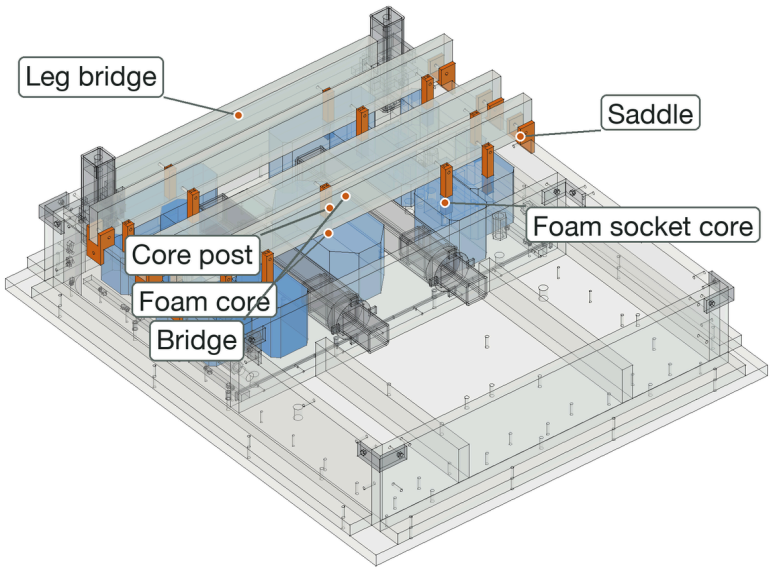


5. Screw the inserts to the floor and the tie-rod couplers onto their greased bolts; hold the draw-rod couplers on bolts through the partition. Slide the spigots through the bushings and put the rear tubes on them, with their sleeves and end plugs. Bolt the leg holders into the corners, drop the legs in, set each at its height and tighten its

pinch screw.

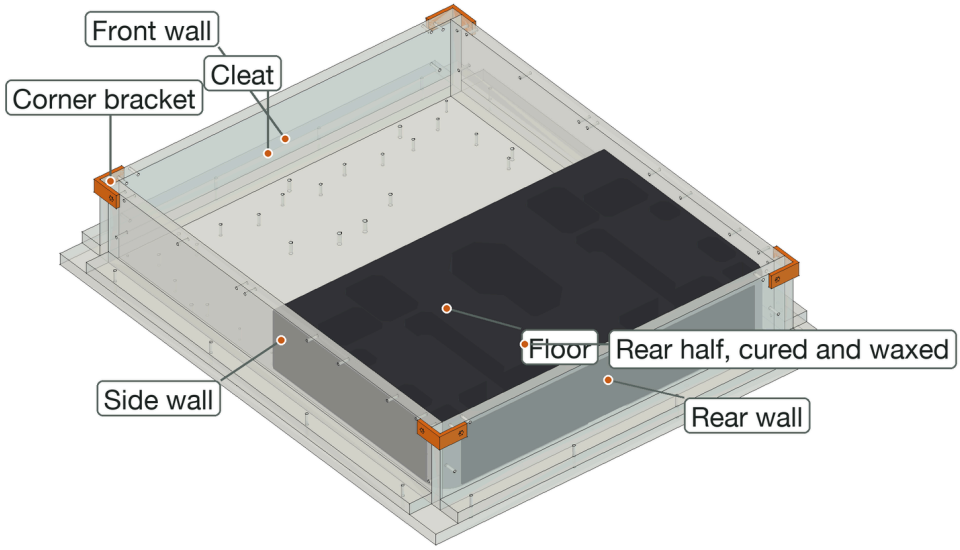


6. Cast the 20 mm skin first. Then set the foam cores on it, drop the bridges into the saddles and bolt each core's post to its bridge. Fill to the pour line.

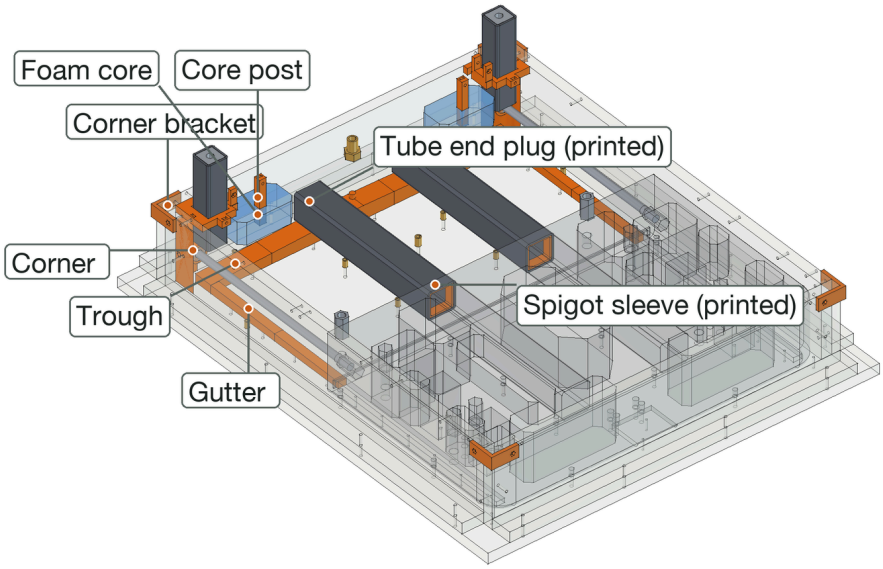


Base, front half (against the rear half)

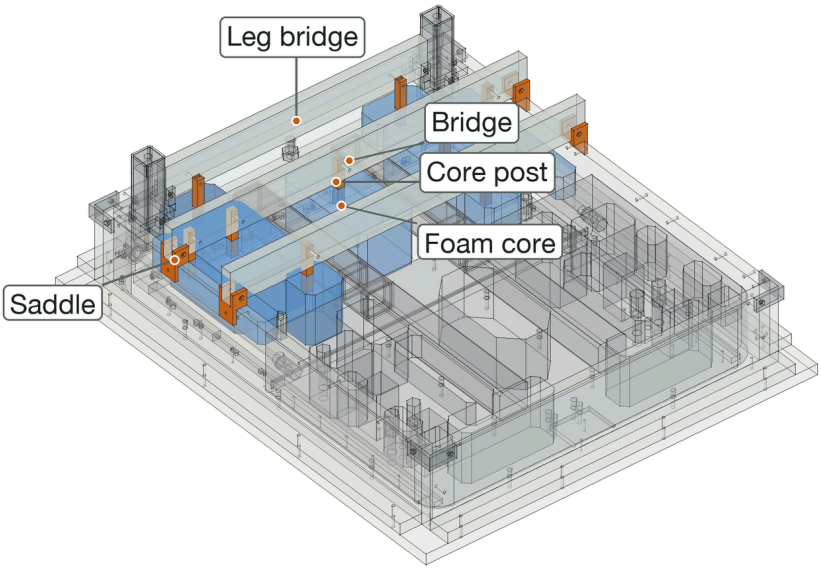
1. After the cure: unbolt the bridges, the partition and its struts, and leave the rest. Dig the foam out of the four sockets, pull the gasket-groove beads, tape over the groove, and wax and PVA the rear half's split face and spigots.



2. Bolt on the trough and gutter formers (the drain liner on the trough's spigot) and the front corner fillers; put the front tubes, sleeves and plugs onto the spigots; screw the draw rods into the rear half's couplers with their ends in the corner fillers; inserts, couplers and legs as before.



3. Skin, foam cores, bridges, fill. After the cure unbolt everything and split the halves with a few wedges at the split, gently.



9. Pouring, and the pour line

- The pour line is 25 mm below the wall tops. Mark it inside each wall with a pencil line, and screed the top with a straight board that has two 25 mm feet riding on the wall tops.
- Vibrate each layer until it stops bubbling; the brackets and cleats hold the walls, so you can vibrate the floor board hard. Check the corner brackets and saddle bolts are still tight halfway through.
- Keep the mix off the bridges and saddles; scrape it off before it cures.

10. Taking it apart, and reuse

1. Leave it the full cure time (build guide). Unbolt the bridges and saddles, then the corner brackets, then the cleats; pull the walls straight off (a plastic wedge in the corner helps).
2. Unscrew every insert screw, coupler bolt and rod from outside, then lift the casting off the floor (or the floor off the casting).
3. Unbolt and pull the formers and fillers (the bolts come out from the board side first). A wooden dowel and a light tap frees a stuck one. Dig out the socket foam; the pocket cores can stay.
4. Scrape the boards, re-wax them, and store the kit pieces with their nuts in a box per mold. The same kit does the next casting; only the formers and pieces that stay in are per casting.

11. If something doesn't fit

Problem	Fix
A nut won't slide under the hole	Clean the pocket with a nut on a bolt; a warm nut (hair dryer) pushes in
A bracket or saddle is loose on the board	Your board is thinner than set: shim with tape, or set `mold_board` and reprint
A pin won't go into its hole	Run a 1/8 in bit through the hole again; trim the pin tip with a knife
A foam core moves while pouring	Check its post bolt is tight and the post is glued; add carpet tape under it
Mix leaks at a corner	A silicone fillet inside the corner before waxing; check the diagonals and brackets
A former is stuck after curing	Bolts out first; then a dowel and a light tap from its thick end